

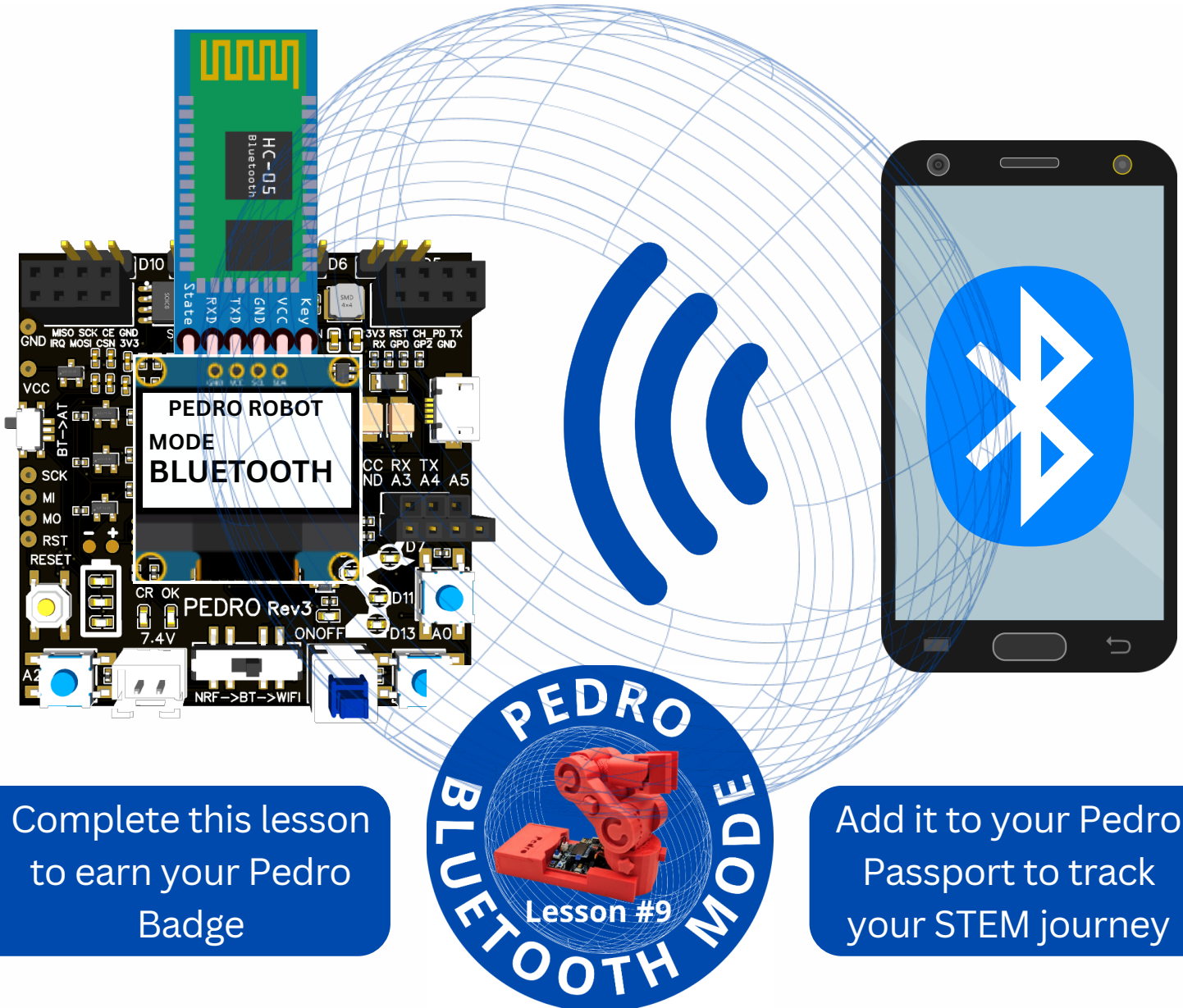
BEGINNER

pedro



STEM Program

Lesson #9: Bluetooth Mode



Complete this lesson
to earn your Pedro
Badge

Add it to your Pedro
Passport to track
your STEM journey

YOUR MISSION: Learn how robots exchange data through a
Bluetooth communication.

Learning Goals

In this lesson, students will discover how to control Pedro wirelessly using Bluetooth technology.

By the end of this lesson, students will be able to:

- Understand Bluetooth communication
- Explain how smartphones communicate with robots
- Connect Pedro to a mobile device



At the end of this lesson, you will unlock your Pedro Badge.

 the Pedro Bluetooth Badge

Teacher Notes

- No prior robotics experience required.
- Duration: 60–90 minutes
- Difficulty: ● Beginner
- Required Materials
 - **1 Assembled Pedro** robot **per student or per group** (*recommended: groups of 2 students*)
 - **HC-05 Bluetooth** module
 - Smartphone with **Serial Bluetooth application** (available on iOS or Android)
 - **OLED Screen 128X64** and **USB cable**
 - **1 computer** with the Arduino IDE installed



Think: The **brain icon** encourages students to make careful observations, share their ideas, and engage in discussion by explaining and justifying their answers.



Learn: The **light bulb icon** indicates important concepts that students should read carefully. Students are encouraged to ask questions and discuss these topics about the robotics concepts presented in the lesson.



Make: The **pencil icon** indicates that students must complete the activity on the page. These activities are designed to reinforce the concepts learned through experimentation, and problem-solving.



Bluetooth Communication

Bluetooth is one of the most common wireless communication technologies. It allows devices to exchange data over short distances without cables.

Bluetooth is used in:

- wireless headphones
- smartwatches
- drones
- robots
- home automation systems



Pedro uses the HC-05 Bluetooth module to create a wireless communication link with a smartphone. The **HC-05 Bluetooth module** connected to the **ATmega32U4** microcontroller through serial communication pins (RX/TX).

The HC-05 module:

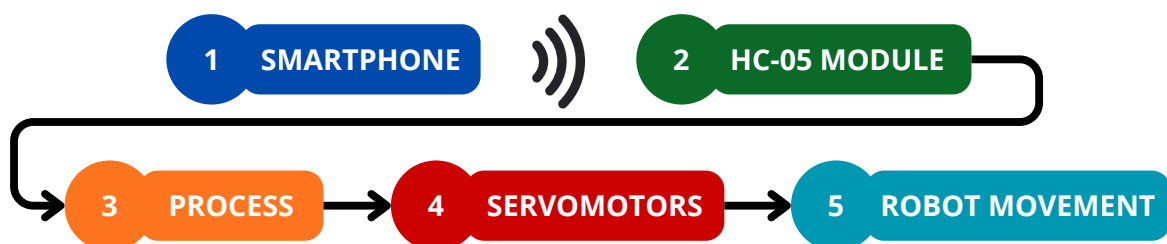
- receives wireless data
- converts Bluetooth signals into serial data
- sends commands to Pedro's program

The HC-05 can operate in two modes:

Mode	Function
Normal Mode	Standard Bluetooth communication
AT Mode	Configuration and settings

Bluetooth communication works like a wireless conversation. The smartphone sends digital commands. Pedro receives the commands and converts them into robot movements.

The Bluetooth Communication Chain is:

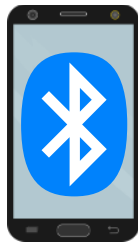




How Bluetooth Mode Works

1

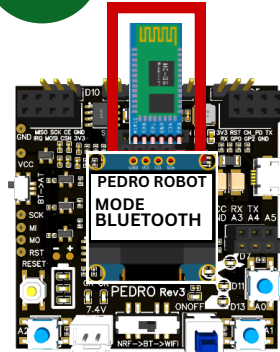
SMARTPHONE



The application sends a text command via Bluetooth

2

HC-05 MODULE

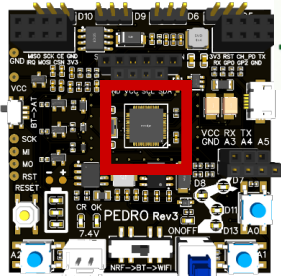


The HC-05 module receives the text command.



3

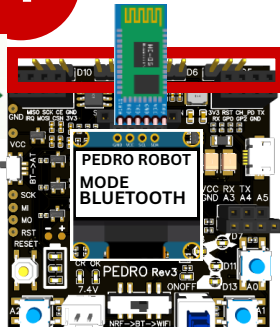
PROCESS



The Pedro's program decodes the command and send PWM signal

4

SERVOMOTORS



The selected servo receives the PWM signal and rotates in the requested direction.

5

ROBOT MOVEMENT



The selected servo rotates, causing the robot to perform the requested movement.



How to Configure Bluetooth Mode

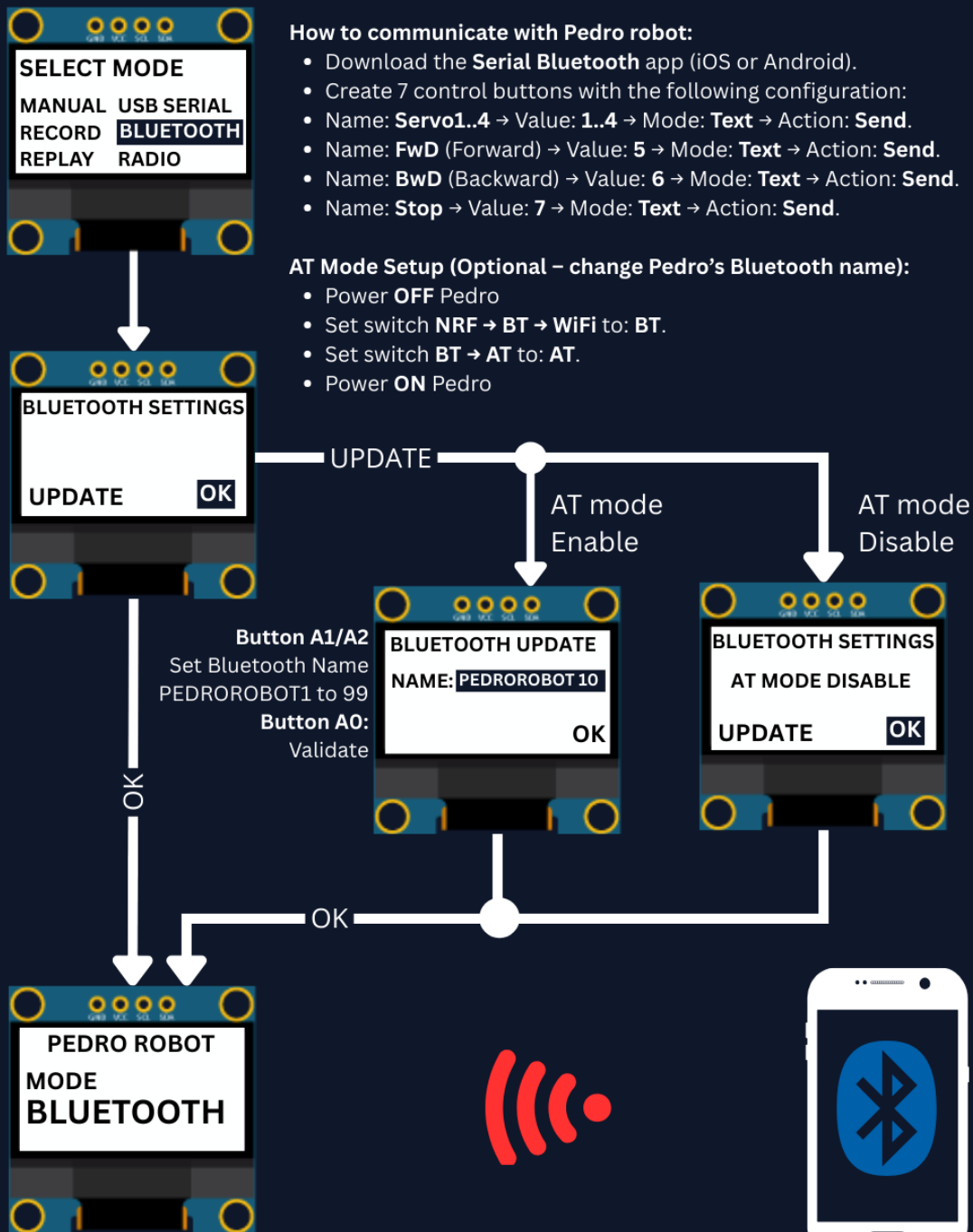
Pedro Bluetooth Mode

How to communicate with Pedro robot:

- Download the **Serial Bluetooth** app (iOS or Android).
- Create 7 control buttons with the following configuration:
- Name: **Servo1..4** → Value: **1..4** → Mode: **Text** → Action: **Send**.
- Name: **FwD** (Forward) → Value: **5** → Mode: **Text** → Action: **Send**.
- Name: **BwD** (Backward) → Value: **6** → Mode: **Text** → Action: **Send**.
- Name: **Stop** → Value: **7** → Mode: **Text** → Action: **Send**.

AT Mode Setup (Optional – change Pedro's Bluetooth name):

- Power **OFF** Pedro
- Set switch **NRF → BT → WiFi** to: **BT**.
- Set switch **BT → AT** to: **AT**.
- Power **ON** Pedro



<https://github.com/almtzr/PedroRobot>



Board Configuration (skip if already configured)

The **Pedro Program** is the core software that powers and controls the robot. It acts as the robot's "brain," managing all behaviors and interactions.

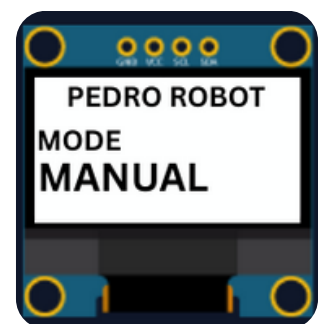
Through this single program, students can access and explore multiple control modes, allowing them to understand how a robot receives instructions, processes them, and translates them into physical actions.

1. **Insert** the Oled Screen into the Pedro board
2. **Connect** the board to your computer with the Micro USB cable.
3. Open **Arduino IDE**
4. **Install the required libraries** from the Library Manager:
 - Tools → Manage Libraries → search **PedroRobot** → Install
 - Tools → Manage Libraries → search **U8glib** → Install
 - Tools → Manage Libraries → search **RF24** → Install
5. Select the port that appear when you connect Pedro robot: **Tools → Port**
6. Select the board: **Tools → Board → Arduino AVR Boards → Arduino Micro**
7. Open the PedroRobot program: **File → Examples → PedroRobot → Pedro**
8. **Compile and upload** the program to your Pedro board.

✓ If everything is correct, Pedro's OLED screen will display **MANUAL MODE**.

Press **A0** and observe what happens. Each time you press **A0**, Pedro selects the next servo.

The LED corresponding to the selected servo lights up, indicating which part of the robot you are controlling. The mapping between the LEDs and the servos is shown below:

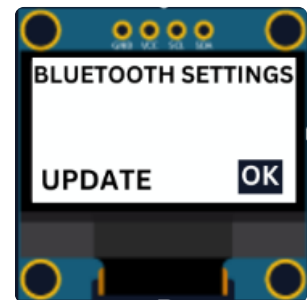


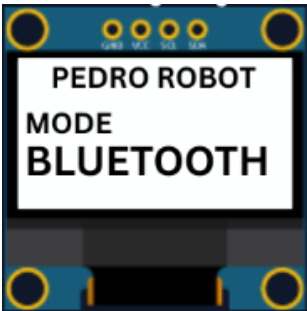
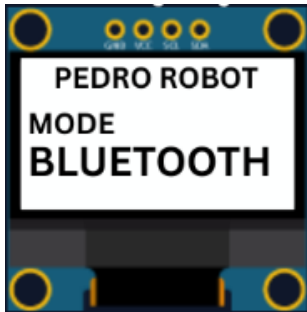
- LED D13 → servo D5 (**Base**)
- LED D11 → servo D6 (**Shoulder**)
- LED D8 → servo D9 (**Elbow**)
- LED D7 → servo D10 (**Gripper**)



Configure Pedro in Bluetooth Mode

1. Power OFF Pedro
2. Set switch **NRF** → **BT** → **WiFi** to **BT**
3. Power ON Pedro (💡 **Note**: maintain **AT** Button to start on **AT mode** for settings options)
4. Enter the **Select Mode** menu (hold **A0** for 4 seconds)
5. Select **BLUETOOTH mode** from the menu and press **A0** to confirm
6. The OLED screen displays:



UPDATE (AT mode required)	OK
7. When you choose UPDATE and press A0 8. The OLED screen displays:  9. That screen allows to update the name of Pedro for Bluetooth connexion 10. Set the Pedro Bluetooth name by using the button A1/A2 PEDROROBOT1 to 99 11. And click on A0 to validate 12. The OLED screen displays: 	7. When you choose OK and press A0 8. The OLED screen displays:  9. 💡 Note: Pedro is now ready to communicate by Bluetooth.
💡 Note: You can now detect the Pedro robot using your smartphone's Bluetooth search. Pedro may appear with the default Bluetooth name HC-05 or with a custom name such as PEDROROBOTn.	



Installing the Smartphone Application

To communicate wirelessly with Pedro, students must install a **Serial Bluetooth application** on their smartphone. These application allows to send commands to the robot through the **HC-05 Bluetooth module**.

Recommended Applications:

Android	iOS
● Serial Bluetooth Terminal ● Bluetooth Electronics	● Bluetooth Serial Controller

After installing the application, students will create a control interface to operate Pedro remotely. The interface uses buttons that send text commands to the robot. Create the following buttons with this configuration:

Button Name	Value Sent	Mode or Type	Action
Servo1	1	Text	Send
Servo2	2	Text	Send
Servo3	3	Text	Send
Servo4	4	Text	Send
FwD (Forward)	5	Text	Send
BwD (Backward)	6	Text	Send
Stop	7	Text	Send

These buttons (**ServoN**) select which servo motor will be controlled and send the movement direction (**FwD** or **BwD**) of the selected servo. When a student presses a button on the smartphone:

1. The app sends a text command via Bluetooth
2. The HC-05 module receives the command
3. The Pedro's program interprets the instruction
4. The selected servo motor moves

Example:

- Button Pressed «**Servo 1**»: Send command «**1**» → the LED **D13** light **ON**
- Button Pressed «**FwD**»: Send command «**5**» → the **Servo Base** Move Forward



Compare Communication Methods

Complete the table.

Feature	UART	Radio	Bluetooth
Wired	☐	☐	☐
Wireless	☐	☐	☐
Very Fast	☐	☐	☐
Long Distance	☐	☐	☐
Requires Pairing	☐	☐	☐



Lesson Summary.

In this lesson, you learned that:

- **UART** is a wired serial communication protocol.
- Communication requires two wires: **TX** and **RX**.
- The Transmitter sends serial data through its **TX pin**.
- The Receiver receives the data through its **RX pin**.
- The receiving microcontroller processes the command and generates the **PWM signal**.
- The selected servo performs the requested movement.



Quiz - Check Your Understanding

1. What module allows Pedro to communicate via Bluetooth?

- ☐ ESP8266
- ☐ NRF24L01
- ☐ HC-05
- ☐ GPS module

2. What type of communication does Bluetooth use?

- ☐ Wired communication
- ☐ Wireless communication
- ☐ Optical communication
- ☐ Mechanical communication

3. What device is commonly used to control Pedro in Bluetooth Mode?

- ☐ Printer
- ☐ Smartphone or tablet
- ☐ Keyboard only
- ☐ Camera

4. What is the role of the HC-05 module?

- ☐ Charge the battery
- ☐ Receive Bluetooth commands
- ☐ Rotate servos directly
- ☐ Print 3D parts

5. What happens when the "FwD" button is pressed in the smartphone app?

- ☐ The robot turns OFF
- ☐ A Bluetooth command is sent to move the servo forward
- ☐ The OLED screen resets
- ☐ The battery charges

6. Which button is used to enter the Select Mode menu?

- ☐ RESET
- ☐ A2
- ☐ A0
- ☐ POWER

7. Which communication protocol is used between the HC-05 and the microcontroller?

- ☐ HDMI
- ☐ Ethernet
- ☐ UART Serial
- ☐ VGA

8. What is the purpose of AT Mode on the HC-05 module?

- ☐ Move the servos
- ☐ Configure Bluetooth settings
- ☐ Charge the robot
- ☐ Record movements

9. What can students modify in AT Mode?

- ☐ Battery voltage
- ☐ OLED size
- ☐ Bluetooth device name
- ☐ Servo dimensions

10. Why is wireless communication important in robotics?

- ☐ It makes robots heavier
- ☐ It allows robots to communicate without cables
- ☐ It changes the robot color
- ☐ It increases battery size

Congratulations Mission Completed!

You have unlocked: 🏆 the Pedro Bluetooth Mode badge
Don't forget to stick your Pedro Badge into your Pedro Passport before starting the next lesson.

